

Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	
Project Name	Smart lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	Followed Python indentation standards.
2	Proper File Structure	Files and folders are logically organize	Yes	Separate folders for templates, static files, models, and database.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as applicant_income, confidence, prediction are meaningful.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like loginUser(), submit(), save_prediction() are descriptive.
5	Code Comments	Inline and block comments explain log	Yes	Important sections are documented with comments.
6	Modular Design	Code is split into reusable functions/m	Yes	Authentication, prediction, PDF generation, and database operations are modular.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code has been optimized by removing duplicate logic.
8	Error Handling	Exceptions and errors are handled gra	Yes	Login validation, session checking, and database validations implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Flask	Login, Register, Dashboard	High
2	Database Module	SQLite	User Management, Prediction History	High
3	Prediction Module	Scikit-learn (Random Forest)	Loan Eligibility Prediction	High
4	PDF Report Module	ReportLab	Report Generation	Medium
5	Dashboard Module	Flask + HTML	Dashboard Analytics & History	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules
Readability	5	Meaningful variable names and consistent formatting
Reusability	5	Modular design allows code reuse across the application
Documentation / Comments	4	Most important functions are documented with comments
Overall Score	4.8	Clean, modular, readable, and maintainable code